Experiment 1

```
void setup() {
  pinMode(2, OUTPUT);
  pinMode(3, OUTPUT);
  pinMode(4, OUTPUT);
  pinMode(5, OUTPUT);
  pinMode(6, OUTPUT);
}
void loop() {
  digitalWrite(2, HIGH);
  delay(1000);
  digitalWrite(2, LOW);
  delay(100);

  digitalWrite(3, HIGH);
  delay(1000);
  digitalWrite(3, LOW);
  delay(100);

  digitalWrite(4, HIGH);
  delay(1000);
  digitalWrite(4, LOW);
  delay(100);

  digitalWrite(5, HIGH);
  delay(1000);
  digitalWrite(5, LOW);
  delay(100);

  digitalWrite(6, HIGH);
  delay(1000);
  digitalWrite(6, LOW);
  delay(100);

  digitalWrite(5, HIGH);
  delay(1000);
  digitalWrite(5, LOW);
  delay(100);

  digitalWrite(4, HIGH);
  delay(1000);
  digitalWrite(4, LOW);
  delay(100);

  digitalWrite(3, HIGH);
  delay(1000);
  digitalWrite(3, LOW);
  delay(100);

  digitalWrite(2, HIGH);
  delay(1000);
}
```

Experiment 2

```
void setup()
{
  pinMode(2, OUTPUT);
}
void loop()
{
  digitalWrite(2, HIGH);
  delay(10000);
  digitalWrite(2, LOW);
  delay(5000);
}
```

Experiment 3

```
// Block-based code
Set input to read ultrasonic distance sensor
    on trigger pin 6, echo pin 5, in units cm
print to serial monitor input with newline
if input <= 100 then
    set pin 8 to HIGH
    set pin 9 to HIGH
    set pin 10 to LOW
else
    set pin 8 to LOW
    set pin 9 to LOW
    set pin 10 to HIGH
```

Experiment 4

```
const int poten = A3;
int var;
void setup()
{
  pinMode(6, OUTPUT);
}
void loop()
{
  var = analogRead(poten);
  int pwmValue = map(var, 0, 1023, 0, 255);
  analogWrite(6, pwmValue);
}
```

Experiment 5

```
int sensorPin = A0;
int sensorValue = 0;
int led = 9;
void setup()
{
  pinMode(led, OUTPUT);
  Serial.begin(9600);
}
void loop()
{
  sensorValue = analogRead(sensorPin);
  Serial.println(sensorValue);
  if (sensorValue < 100)
  {
    Serial.println("LED light on");
    digitalWrite(led, HIGH);
    delay(1000);
    digitalWrite(led, LOW);
  }
  else
  {
    digitalWrite(led, LOW);
  }
  delay(200);
}
```

Experiment 6

```
int moistureValue;
float moisture_percentage;
void setup()
{
  pinMode(7, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(5, OUTPUT);
  Serial.begin(9600);
}
void loop()
{
```

```
  moistureValue = analogRead(A0);
  moisture_percentage = (moistureValue * 100.0) / 539.0;
  if (moisture_percentage >= 0 && moisture_percentage < 25)
  {
    digitalWrite(7, HIGH);
    digitalWrite(6, LOW);
    digitalWrite(5, LOW);
  }
  else if (moisture_percentage >= 25 && moisture_percentage < 80)
  {
    digitalWrite(7, LOW);
    digitalWrite(6, HIGH);
    digitalWrite(5, LOW);
  }
  else if (moisture_percentage >= 80 && moisture_percentage <= 100)
  {
    digitalWrite(7, LOW);
    digitalWrite(6, LOW);
    digitalWrite(5, HIGH);
  }
  Serial.print("Moisture Value : ");
  Serial.print(moisture_percentage);
  Serial.println("%");
  delay(1000);
}
```

Experiment 7

```
#include <LiquidCrystal.h>
const int rs = 13, en = 12, d4 = 11, d5 = 10, d6 = 9, d7 = 8;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
int Contrast = 0;
void setup()
{
  Serial.begin(9600);
  analogWrite(6, Contrast);
  lcd.begin(16, 2);
  lcd.setCursor(4, 0);
  lcd.print("pH Value:");
}
void loop()
{
  int sensorValue = analogRead(A0);
  float ph = sensorValue * (14.0 / 1023.0);
  Serial.println(ph);
  lcd.setCursor(0, 1);
  lcd.print("          ");
  lcd.setCursor(0, 1);
  lcd.print(ph, 2);
  lcd.print(" ");
  if (ph >= 0.0 && ph < 5.0)
  {
    lcd.print("ACID");
  }
  else if (ph >= 5.0 && ph < 7.0)
  {
    lcd.print("Normal");
  }
  else if (ph >= 7.0 && ph <= 14.0)
  {
    lcd.print("Base");
  }
  else
  {
    lcd.print("Invalid");
  }
  delay(1000);
}
```

Experiment 8

```
// Block-based code
set Gas_sen to read analog pin A0
print to serial monitor Gas_sen with newline
configure LCD 1 type to I2C (MCP23008) with address 32 (0x20)
set position on LCD 1 to column 0 row 0
print to LCD 1 "SENSOR ACTIVE"
if Gas_sen >= 470 then
    set position on LCD 1 to column 0 row 1
    print to LCD 1 "Detected"
else if Gas_sen >= 2 then
    set position on LCD 1 to column 0 row 1
    print to LCD 1 "Not Detected"
else
    set position on LCD 1 to column 0 row 1
    print to LCD 1 "Not Detected"
```

Experiment 10

```
const int MAX_LEN = 30;
char sendMsg[MAX_LEN] = "Hello i'm Arduino-2\n";
char receiveMsg[MAX_LEN];
int switchState = 1; // Active HIGH input
int switch_pin = 10;

void setup() {
  Serial.begin(9600);
  pinMode(switch_pin, INPUT);
}

void loop() {
  switchState = digitalRead(switch_pin);
  if (switchState == HIGH) {
    // Receive if switch is off
    if (Serial.available() > 0) {
      int len = Serial.parseInt();   // Read the length as number
      Serial.read();                 // Consume the newline after the number

      int n = Serial.readBytes(receiveMsg, len);
      receiveMsg[n] = '\0';          // Null-terminate the received string

      Serial.print("Message = ");
      Serial.println(receiveMsg);
    }
  } else {
    // Send if switch is on
    int len = strlen(sendMsg);
    Serial.write(receiveMsg, len); // send raw message
    delay(1000);
  }
}
```

Experiment 11

```
int trigPin = 9;
int echoPin = 10;
int a = 2, b = 3, c = 5, d = 6, e = 4;
long dist;
long ival;
void setup()
{
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(a, OUTPUT);
  pinMode(b, OUTPUT);
  pinMode(e, OUTPUT);
  pinMode(c, OUTPUT);
  pinMode(d, OUTPUT);
  Serial.begin(9600);
```

```
}
void loop()
{
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  ival = pulseIn(echoPin, HIGH);
  dist = (ival / 2) / 29.154;
  Serial.print("dist:");
  Serial.print(dist);
  Serial.println("CM");
  if (dist <= 50) {
    digitalWrite(a, HIGH);
    digitalWrite(b, LOW);
    digitalWrite(e, LOW);
    digitalWrite(c, LOW);
    digitalWrite(d, LOW);
  }
  else if (dist <= 100) {
    digitalWrite(a, LOW);
    digitalWrite(b, HIGH);
    digitalWrite(e, LOW);
    digitalWrite(c, LOW);
    digitalWrite(d, LOW);
  }
  else if (dist <= 150) {
    digitalWrite(a, LOW);
    digitalWrite(b, LOW);
    digitalWrite(e, HIGH);
    digitalWrite(c, LOW);
    digitalWrite(d, LOW);
  }
  else if (dist <= 200) {
    digitalWrite(a, LOW);
    digitalWrite(b, LOW);
    digitalWrite(e, LOW);
    digitalWrite(c, HIGH);
    digitalWrite(d, LOW);
  }
  else if (dist <= 250) {
    digitalWrite(a, LOW);
    digitalWrite(b, LOW);
    digitalWrite(e, LOW);
    digitalWrite(c, LOW);
    digitalWrite(d, HIGH);
  }
  else
  {
    digitalWrite(a, HIGH);
    digitalWrite(b, HIGH);
    digitalWrite(e, HIGH);
    digitalWrite(c, HIGH);
    digitalWrite(d, HIGH);
  }
  delay(50);
}
```

Experiment 12

```
#include <Adafruit_LiquidCrystal.h>
#include <Keypad.h>

// LCD 1 - Shows key pressed
Adafruit_LiquidCrystal lcd_1(0x20);

// LCD 2 - Shows total key count
Adafruit_LiquidCrystal lcd_2(0x21);
const byte ROWS = 4;
```

```
const byte COLS = 4;

char hexaKeys[ROWS][COLS] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'},
};

byte rowPins[ROWS] = {9, 8, 7, 6};
byte colPins[COLS] = {5, 4, 3, 2};

Keypad customKeypad = Keypad(makeKeymap(hexaKeys), rowPins,
colPins, ROWS, COLS);

int keyPressCount = 0;
void setup() {
  Serial.begin(9600);

  // Initialize both LCDs
  lcd_1.begin(16, 2);
  lcd_1.setBacklight(1);
  lcd_1.print("Enter key value");
  lcd_1.setCursor(0, 1);

  lcd_2.begin(16, 2);
  lcd_2.setBacklight(1);
  lcd_2.print("Total Key Press:");
  lcd_2.setCursor(0, 1);
  lcd_2.print("Count: 0");
}
void loop() {
  char customKey = customKeypad.getKey();
  if (customKey) {
    Serial.println(customKey);

    // Display key on LCD 1
    lcd_1.setCursor(7, 1);
    lcd_1.print(" ");
    lcd_1.setCursor(7, 1);
    lcd_1.print(customKey);

    // Increment and show count on LCD 2
    keyPressCount++;
    lcd_2.setCursor(0, 1);
    lcd_2.print("Count: ");
    lcd_2.print(keyPressCount);
    lcd_2.print(" "); // Clear extra digits if count goes down

    delay(100);
  }
}
```